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# EVALUATING FEATURE PREDICTION FOR 2D POSE TRAJECTORY RESTORATION WITH PAIRED SYNTHETIC SUPERVISION

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Paper under double-blind review

## ABSTRACT

Can predicting reference features help restore movement changes without distorting their trajectories or anatomical side? We study this question using paired synthetic motion, projected reference poses, and controlled observation and naming errors. Eight model families and a supervision follow-up are evaluated on the same 14 development people and three fitted seeds. Direct coordinate training improves noisy-pose response error from  $12.69^\circ$  to  $7.54^\circ$ , yet predicting zero change scores  $5.81^\circ$ , below every neural variant in the pooled response experiment. Feature-difference pretraining has an uncertain  $0.37^\circ$  advantage over endpoint-feature supervision (95% interval  $[-1.11, 1.76]^\circ$ ), with reversed ordering under coordinate-only readouts and unequal initial auxiliary influence. The original change-supervised package worsens all eight waveform means; a follow-up on two frozen encoders identifies substantial sensitivity to scalar-loss weighting. A post hoc diagnostic also finds worse geometric side assignment for the feature-difference variant despite its lower response point estimate. These comparisons show how supervision and failure accounting constrain evidence of faithful movement restoration, while leaving the incremental benefit of the tested predictive representations unresolved.

## 1 INTRODUCTION

Restoring a pose sequence is often an intermediate step toward measuring movement. A system may compare a person’s walking before and after an intervention, or compare the excursion of the two knees. In these settings, an output that is closer to observed or reference coordinates can still attenuate a movement change or attach it to the wrong anatomical side. The representation must support the measurement after decoding, and the decoder must retain the structure on which that measurement depends.

Feature prediction offers a plausible route to this goal. Learning to predict a reference representation could emphasize movement structure rather than individual coordinate errors. Yet this hypothesis concerns a downstream measurement, not merely the predictability of latent features. A frozen representation, a supervised coordinate readout, and a scalar change objective each impose different constraints. Their combination needs to be evaluated against alternatives that reveal whether an apparent gain represents useful restoration.

We examine that connection in a controlled 2D setting. Each source movement supplies paired states with a known synthetic knee edit. Rendering and pose estimation produce noisy observations; separately applied naming swaps make anatomical left–right labeling an explicit challenge. The original research questions ask whether reference-feature prediction improves a signed bilateral movement response and whether change supervision improves the paired measurement without damaging trajectories. We subsequently inspect geometric side assignment to test whether the response ranking agrees with naming fidelity.

The strongest finding is a constraint on interpreting the learned systems. Improvements over noisy observations coexist with a better pooled score from predicting no change. The delta–endpoint feature comparison remains uncertain, and the readout package changes waveform quality substantially. Our contributions are a controlled paired evaluation linking response and restoration fidelity,

a comparison that exposes the limits of the tested feature-prediction advantage, and a supervision follow-up with explicit reliability and post hoc laterality analysis. The same development population informs all stages; the study supplies diagnostic evidence, not independent confirmation of a new representation method.

## 2 POSITIONING RELATIVE TO SKELETAL PREDICTION AND RESTORATION

Joint-embedding prediction learns representations by predicting missing information in a target feature space, as in I-JEPA and video feature prediction (Assran et al., 2023; Bardes et al., 2024). The closest skeletal precedent is S-JEPA (Abdelfattah & Alahi, 2024), which predicts masked skeletal features and evaluates action recognition. General Feature Prediction likewise studies learned targets for masked skeleton modeling (Sun et al., 2025). Our S-JEPA-inspired procedure changes both the information available to the teacher and the downstream task: estimated 2D poses enter the student, projected reference poses enter a privileged teacher, and the deployed student supports coordinate restoration. S-JEPA instead uses its target encoder downstream. We therefore evaluate this adaptation rather than reproduce a recognition benchmark.

Temporal restoration also has established reconstruction-based approaches. PoseBERT learns temporal 3D modeling through masked modeling and denoising of motion-capture sequences (Baradel et al., 2022); MotionBERT reconstructs 3D motion from noisy and partial 2D observations to learn transferable representations (Zhu et al., 2023). SmoothNet refines temporal pose estimates to reduce jitter (Zeng et al., 2022). These works motivate reconstruction and refinement controls, but their reported tasks do not settle the particular paired response comparison examined here. None was fitted as an external baseline in this study. The contribution is consequently an evaluation of the implemented procedures, with response, trajectory, and side assignment assessed together, rather than a claim of superiority over these methods.

## 3 PAIRED MOVEMENT RESTORATION

### 3.1 STATES, PROJECTED TARGETS, AND ANATOMICAL NAMES

Let  $M_a$  denote a complete body-model motion window and  $M_b = \mathcal{E}_\delta(M_a)$  its edited counterpart. The nominal right-knee edit  $\delta \in \{0, 5, 10, 15\}^\circ$  precedes optional physical mirroring  $T_m$ . A fixed camera  $c$  gives the 2D reference  $Y_i = \pi_c(T_m M_i)$  for each state  $i \in \{a, b\}$ . Rendering, a fixed pose estimator, added occlusion, and a naming transformation produce observation  $X_i$ , including coordinates, confidence, availability, and timestamps. Paired states share their camera, physical orientation, estimator, occlusion, and naming conditions. The restoration model predicts  $\hat{Y}_i$  from one complete observed window.

For anatomical leg  $\ell \in \{L, R\}$ ,  $\theta_{i,\ell,t}$  is the image-plane knee angle between knee-to-hip and knee-to-ankle vectors. Define

$$q_{i,\ell} = P_{95}(\theta_{i,\ell}) - P_5(\theta_{i,\ell}), \quad A_i = q_{i,R} - q_{i,L}, \quad \Delta A = A_b - A_a. \quad (1)$$

The response is a between-state contrast, not a temporal derivative. It is side-sensitive, but does not uniquely identify which leg changed. The nominal 3D edit need not equal the projected response; all targets are recomputed after projection.

Exchanging already-computed excursions  $(q_L, q_R)$  reverses  $A = q_R - q_L$  algebraically. Physical mirroring is a different operation: it reflects 3D geometry, exchanges body-model sides, and reprojects through the same camera. The resulting projected excursions need not exchange exactly, so we impose no sign-equivariance assumption. A global or temporary naming swap changes only the estimated input, including confidence and availability. The anatomically named reference remains fixed. This separation makes naming errors a restoration problem rather than an intended movement response.

### 3.2 REPRESENTATIONS, READOUTS, AND COMPARISON LOGIC

Four-frame patches from 12 joints give 384 tokens. A bidirectional transformer uses the full window; this is offline restoration rather than future prediction. Masked feature pretraining predicts

### A States, references, and observations

$$M_a \xrightarrow{\text{knee edit } \mathcal{E}_\delta} M_b = \mathcal{E}_\delta(M_a)$$

$$Y_i = \pi_c(T_m M_i)$$

$$X_i = N_n[\text{Pose}(\text{Render}_{c,o}(T_m M_i))]$$

$i \in \{a, b\}$ ; paired camera / observation condition

$T_m$ : physical 3D mirror; reproject  $Y_i$

$N_n$ : input naming error;  $Y_i$  unchanged

### B Feature comparison during training

$$X_i \longrightarrow \text{student + predictor} \longrightarrow p_i$$

$$Y_i \longrightarrow \text{EMA teacher} \longrightarrow t_i$$

$$e_i = H[p_i/\tau_s - \text{sg}((t_i - c)/\tau_t)]$$

$$\text{Endpoint: } (\|e_a\|^2 + \|e_b\|^2)/(2D)$$

$$\text{Delta: } \|e_b - e_a\|^2/(2D)$$

Common masked queries; same base feature loss.

### C Inference

$$X_i \longrightarrow \text{frozen encoder + residual readout} \longrightarrow \hat{Y}_i$$

One observed state; no reference, paired state, teacher, or predictor.

Figure 1: Paired supervision and single-state inference. A:  $T_m$  acts on 3D motion before fixed-camera projection;  $N_n$  exchanges input names while leaving reference names unchanged. Occlusion enters rendering. B: student predictor logits  $p_i$  and detached reference-teacher logits  $t_i$  define residuals at common masked queries;  $H$  removes the channel mean,  $c$  is the teacher center, and  $D = 96$ . Both auxiliaries add to the same base objective; their coefficient and reductions are defined in the text and Appendix A. EMA denotes exponential moving average of student weights. C: pretraining’s teacher and predictor are discarded. The frozen path fits a residual readout; direct coordinate training instead updates encoder and readout jointly. No reconstructed pose examples are implied.

an exponential-moving-average teacher’s reference features with centered, sharpened cross-entropy (Caron et al., 2021) and a VICReg regularizer (Bardes et al., 2022). Normalization uses only retained observations and is shared with teacher references. The target is privileged synthetic supervision, not a second unlabeled observation.

For a common queried token, let  $p_i$  and  $t_i$  be predictor and teacher logits,  $c$  the running teacher center, and  $H$  channel-mean removal. The endpoint and delta auxiliaries are

$$e_i = H[p_i/0.1 - \text{sg}\{(t_i - c)/0.06\}], \quad L_{E,k} = \frac{\|e_a\|^2 + \|e_b\|^2}{2D}, \quad L_{\Delta,k} = \frac{\|e_b - e_a\|^2}{2D}. \quad (2)$$

Losses average supported common queries within pairs, then pairs equally. Both retain the base feature loss and use one training-calibrated coefficient. The delta objective allows common residuals to cancel; it need not improve endpoint reconstruction. Separately evolving teachers and unequal initial auxiliary influence further restrict an isolated coupling interpretation (Section 4.2).

The readout predicts coordinate corrections at observed positions and absolute coordinates at missing positions. Coordinate supervision minimizes  $L_x$ . The original change package is  $L_x + L_s + L_g$ , where  $L_s = \mathbb{E}[(\Delta\hat{A} - \Delta A)/180]^2$  and  $L_g$  penalizes predicted segments shorter than two pixels on fixed reference support. It adds two terms, so its contrast with  $L_x$  cannot isolate the scalar term. The repair keeps coordinate and geometry supervision fixed, comparing  $L_x + 0.1L_s + L_g$  with  $L_x + \lambda_d L_d + L_g$ . Here  $L_d$  penalizes the squared paired angular-change residual at every supported leg and timestamp, normalized by 180<sup>2</sup>. Its initial readout-gradient magnitude is matched to the low-scalar term using training-only batches. Appendix A specifies all reductions and updates.

### 3.3 DATA, OUTCOMES, AND INFERENTIAL SCOPE

AMASS supplies recorded motion-capture sequences (Mahmood et al., 2019). The completed study fits on 112 people, 692 motions, and 1,645 windows; development uses 14 different people and 155

Table 1: What each comparison changes. Every frozen family fits a fresh residual readout. All coordinate targets are projected references. “Change” means the original scalar-plus-geometry package. Equal update totals do not match encoder adaptation or coordinate-supervised exposure.

| Family                        | Pretraining target                                        | Encoder in readout | Supervision / question                               |
|-------------------------------|-----------------------------------------------------------|--------------------|------------------------------------------------------|
| Direct                        | None                                                      | Jointly fitted     | Coordinate or change; practical adaptive restoration |
| Initialized                   | None                                                      | Frozen             | Coordinate or change; untrained-feature control      |
| Coordinate / coordinate delta | Masked reference coordinates; optionally paired residuals | Frozen             | Coordinate or change; reconstruction pretraining     |
| Core / shuffled JEPa          | Matched / shuffled reference features                     | Frozen             | Coordinate or change; correspondence control         |
| Endpoint / delta JEPa         | Base features + endpoint / paired auxiliary               | Frozen             | Coordinate or change; incremental feature comparison |
| Repair                        | Reuse endpoint / delta encoders                           | Frozen             | Low scalar or dense change; readout sensitivity      |

windows. Twelve development people are from BioMotionLab\_NTroje and two from KIT. Windows contain 128 timestamps at 25 Hz. The design crosses two cameras, original/mirrored states, clear/occluded images, three naming conditions, and RTMPose-M, HRNet-W32, and ViTPose-base estimates (Jiang et al., 2023; Sun et al., 2019; Xu et al., 2022). ViTPose and the 15° edit are excluded from fitting. Rendering-derived person boxes remove detector uncertainty from this comparison. Synthetic edits and view-dependent 2D angles do not constitute clinical ground truth.

The completed sequence comprises the core comparison, a feature-response extension, and a readout repair, all using seeds 17, 29, and 43. Each follows inspection of the preceding development results. “Declared primary” means designated in the retained study artifacts; no independent preregistration is asserted. The core primary compares core JEPa with direct fitting under the change package. The response primary compares delta with endpoint JEPa under that package. The repair primary compares dense with low scalar for the delta encoder on ViTPose waveform error. Coordinate-only feature contrasts are secondary; the assignment audit, stratifications, penalty sensitivity, and new figure intervals are exploratory and unadjusted.

Response error is  $|\Delta\hat{A} - \Delta A|$  in degrees. Waveform error measures mean absolute knee-angle error over reference-admitted timestamps and both legs. Coordinate error (NLE) is joint distance divided by the rendered person-box diagonal. Angular eligibility requires at least 16 frames and 80% coverage on common reference support across source-family variants. Predicted nonfinite or too-short segments count as measurement failures, retained with waveform cost 180° and response cost 720°. Geometric assignment uses a separate denominator, defined in Section 4.5. A zero-response predictor supplies a measurement benchmark and has no restored waveform.

Scores average conditions within windows, windows within source motions, motions within people, then seeds. Renderings and windows do not enlarge the independent population. Core and response intervals use 2,000 paired crossed bootstrap draws over people and seeds. Repair’s declared interval uses paired person means after seed averaging and a Student-*t* interval over 14 people, conditional on those fitted seeds. We retain that distinction throughout. Appendix B documents weighting and complete inventories.

## 4 RESULTS: WHAT SUPPORTS FAITHFUL RESTORATION?

### 4.1 COORDINATE RESTORATION HELPS, BUT POOLED RESPONSE RECOVERY IS LIMITED

Direct coordinate training improves response error from 12.69° for unchanged observations to 7.54°, waveform error from 18.57° to 12.07°, and NLE from 0.0718 to 0.0298. Its exploratory paired response gain is 5.14° [2.15, 8.65]° (crossed 95% interval). Every person’s seed-averaged waveform and coordinate errors improve, while response error improves for 10 of 14 people. These results

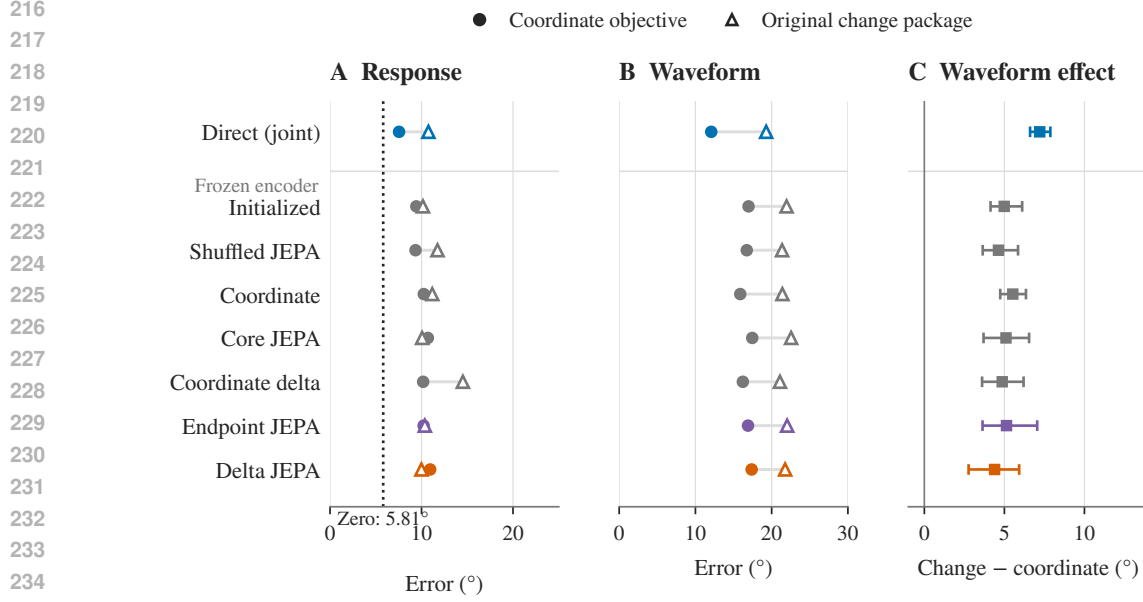


Figure 2: Restoration quality and supervision tradeoffs. A–B: all eight families and both objectives, as hierarchical means over 14 people and three seeds, pooled over estimators. Circles use coordinate supervision; open triangles add scalar and geometry terms. Direct fitting adapts its encoder; all other encoders are frozen. Zero response ( $5.81^\circ$ ) appears only in A. C: paired change-minus-coordinate waveform effects with crossed-person/seed 95% intervals (2,000 draws), newly computed and exploratory; positive values indicate deterioration. Responses and waveforms retain failures at  $720^\circ$  and  $180^\circ$ , respectively. Families share the development cohort and are not independent replications.

demonstrate useful coordinate restoration under the protocol, but are insufficient to establish useful recovery of paired movement change.

Predicting  $\Delta\hat{A} = 0$  gives response error  $|\Delta A|$  and a pooled mean of  $5.811^\circ$ , below all 16 neural variants in the response export. The comparison is especially informative because zero prediction recovers no movement change. The metric rewards avoiding erroneous estimated changes when the true projected response is small, while the large failure cost also penalizes rare invalid reconstructions. A better score than noisy observations therefore cannot by itself support a movement-recovery claim. This observation concerns the pooled task and cost rule, not an assertion that the trained models contain no response information.

An exploratory stratification locates part of this limitation (Table 2). Direct coordinate training beats zero response in both clear-image edit groups, but not in either occluded group. Both predictive change-readout variants remain above zero in all four groups. The  $5/10^\circ$  and  $15^\circ$  labels are nominal edit levels, not bins of true projected response. Per-pair true magnitudes are absent from the compact exports; binning already averaged signed responses would not recover that analysis. We report every selected observation-by-edit group and, in Appendix D, all observation-by-estimator groups and all 16 methods.

#### 4.2 THE INCREMENTAL FEATURE-PREDICTION BENEFIT REMAINS UNRESOLVED

The declared delta–endpoint response comparison is  $9.99^\circ$  versus  $10.36^\circ$ , a  $0.37^\circ$  advantage for delta with crossed 95% interval  $[-1.11, 1.76]^\circ$ . Under coordinate-only readouts the order reverses: delta scores  $10.94^\circ$  and endpoint  $10.23^\circ$ . The secondary readout interaction is  $1.09^\circ$   $[-0.65, 2.85]^\circ$ . Thus the data establish neither the primary benefit nor its apparent dependence on readout choice. The earlier core primary is also unresolved: core JEPA versus direct fitting under the change package gives  $0.69^\circ$   $[-0.64, 1.98]^\circ$ . It does not compare against the stronger direct-coordinate system.

Table 2: Exploratory response errors by observation and nominal edit, in degrees. Direct uses coordinate supervision and joint fitting; endpoint and delta use frozen encoders with the original change readout. All groups contain the same 14 people and three seeds. The last column is a paired zero-minus-direct effect with crossed 95% interval; positive favors direct. Original 720° failure scoring is retained.

| Observation, edit | Zero | Direct | Endpt. | Delta | Zero – direct [95% CI] |
|-------------------|------|--------|--------|-------|------------------------|
| Clear, 5/10°      | 4.10 | 2.79   | 4.54   | 4.33  | 1.32 [1.22, 1.42]      |
| Clear, 15°        | 9.23 | 6.12   | 9.95   | 9.64  | 3.10 [2.88, 3.30]      |
| Occluded, 5/10°   | 4.10 | 9.54   | 12.65  | 12.08 | -5.44 [-8.31, -3.03]   |
| Occluded, 15°     | 9.23 | 14.49  | 17.82  | 17.47 | -5.26 [-8.33, -2.73]   |

The feature contrast additionally compares unequal initial auxiliary influence. The shared coefficient  $\lambda_J = 0.0126468$  was calibrated on 32 fixed training batches to set the larger auxiliary’s root-mean-square parameter-gradient magnitude to 10% of the base objective. This gives the endpoint auxiliary 10%, but the delta auxiliary only 0.0013%, before clipping. Across the six new delta/endpoint pretraining fits, the combined gradient is clipped at norm one on 11,999 of 12,000 updates. These quantities are verified from the calibration receipt and training ledger (Appendix A.5). They do not show that the initial ratio persists or explain the final ordering causally. They do show that sharing a coefficient did not create a comparison with matched initial influence. The uncertain result bounds these fitted procedures rather than predictive representations in general.

#### 4.3 RARE FAILURES MATERIALLY AFFECT THE RESPONSE CONTRAST

With  $F$  denoting a failed eligible response and  $w$  the hierarchical population weights, the score at cost  $C$  is

$$S(C) = \underbrace{\mathbb{E}_w[1_{-F}|\Delta\hat{A} - \Delta A|]}_{U: \text{unconditional successful contribution}} + C \Pr_w(F). \quad (3)$$

At the original  $C = 720^\circ$ , endpoint and delta fail on 0.563% and 0.525% of the weighted population, contributing  $4.055^\circ$  and  $3.778^\circ$  to their scores. Their successful contributions are  $6.306^\circ$  and  $6.210^\circ$ . The  $0.277^\circ$  difference in failure costs accounts for approximately 74% of the  $0.373^\circ$  mean contrast. This is an additive score decomposition, not a causal explanation or a 74% failure rate. Conditional successful-output errors have another denominator:  $U/(1 - \Pr_w(F))$  gives  $6.342^\circ$  and  $6.242^\circ$ , respectively, and cannot be added directly to the failure contribution.

The cost scale follows the outcome range: projected interior angles and excursions lie in  $[0, 180]^\circ$ , so  $A \in [-180, 180]^\circ$  and  $\Delta A \in [-360, 360]^\circ$ ; two valid responses can differ by  $720^\circ$ . We rescore the same predictions at costs 0, 180, 360, and  $720^\circ$ , retaining the original primary. The  $360^\circ$  stress test is also the maximum error of zero prediction;  $180^\circ$  is an angle-error scale, not a response-error bound. Delta’s mean advantage is 0.10, 0.17, 0.23, and  $0.37^\circ$ , respectively, with all intervals spanning zero (Figure 3). No neural variant beats zero response at  $360^\circ$  or  $720^\circ$ ; one does at  $180^\circ$  and two at zero cost. We select no cost as preferable. In particular, both predictive variants’ successful contributions already exceed  $5.811^\circ$ : neither can beat zero response at any nonnegative failure cost with these predictions and weights. Their pooled deficit therefore cannot be attributed to the failure penalty alone. This lower-bound argument assigns failures zero error; it is not a success-only comparison.

#### 4.4 READOUT WEIGHTING SUBSTANTIALLY CHANGES WAVEFORM FIDELITY

The original change package worsens waveform means in all eight model families (Figure 2). In the five core families, deterioration also occurs for every person’s seed-averaged score and every seed’s population mean. Core JEPa illustrates the conflict: response error improves from  $10.69^\circ$  to  $10.07^\circ$ , while waveform error rises from  $17.45^\circ$  to  $22.55^\circ$ . This result concerns the tested scalar-plus-geometry package. It does not isolate which addition causes the deterioration.

The follow-up reuses the delta and endpoint encoders and holds coordinate and geometry supervision fixed (Figure 4). For delta on ViTPose, the original, low-scalar, and dense waveform means are  $23.17^\circ$ ,  $19.47^\circ$ , and  $19.19^\circ$ . Reducing scalar weight improves the mean by  $3.70^\circ$  [2.58,  $4.81^\circ$ ] (exploratory paired-person interval). The declared dense-versus-low-scalar gain is only  $0.28^\circ$  [ $-0.19$ ,

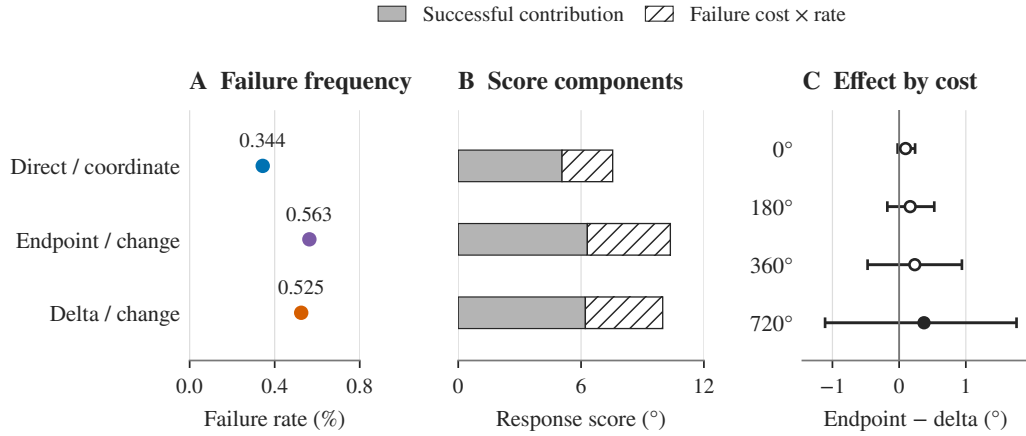


Figure 3: Reliability and uncertainty have different denominators. A: descriptive person-balanced response failure percentages for three fitted procedures. B: unconditional successful-error and 720° failure-cost contributions, whose sum is the response score. Bars are not conditional success-only errors. C: paired endpoint-minus-delta response effects at four fixed costs, with crossed-person/seed 95% intervals over 14 people and three seeds; positive favors delta. The filled point retains the declared 720° comparison. Costs 360° and 180° are exploratory stress tests; zero is an accounting lower bound assigning failures no error. All four intervals cross zero. Methods in A–B are grouped by their named supervision, not matched encoder adaptation.

0.75]°; its crossed-bootstrap sensitivity interval is  $[-0.44, 0.86]^\circ$ . The corresponding response gain is  $0.13^\circ [-2.69, 2.94]^\circ$ , establishing neither superiority nor preserved response accuracy.

The secondary endpoint waveform gain is  $0.72^\circ [0.39, 1.05]^\circ$ , unadjusted and distinct from the delta primary. Both dense arms remain about six degrees worse than the direct-coordinate reference on ViTPose. Low scalar recovers 93% of delta’s mean original-to-dense waveform improvement, but that ratio is subordinate to the absolute effects and does not allocate mechanistic credit. Dense supervision changes temporal information and gradient locations as well as the target; these fits do not identify percentile-gradient sparsity as the cause. The follow-up supports sensitivity to downstream weighting more clearly than a unique advantage of dense supervision.

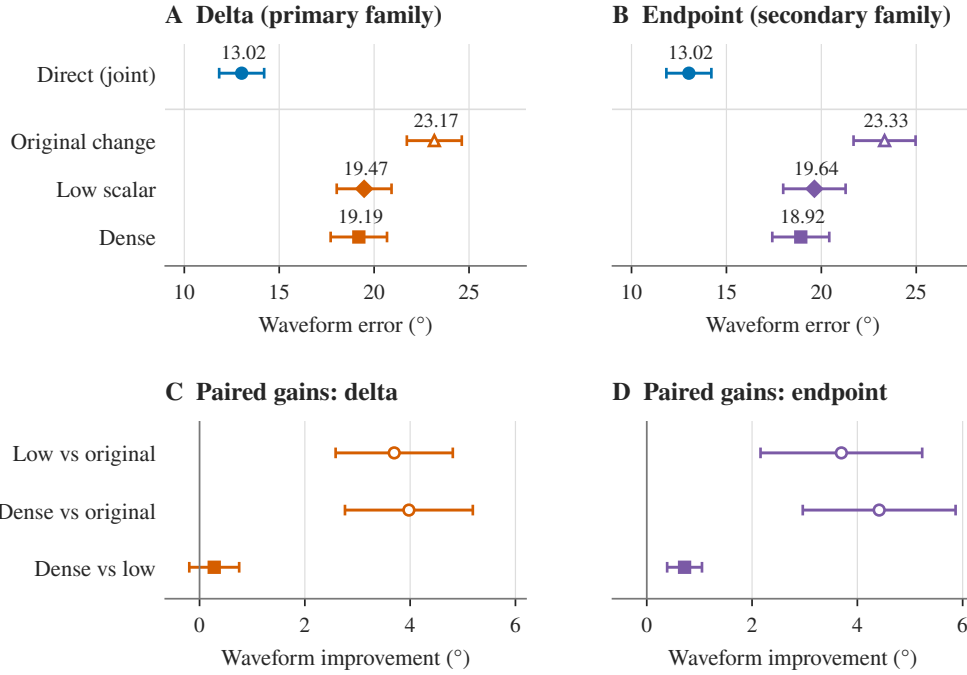


Figure 4: Readout repair on ViTPose. A–B: waveform means and person- $t$  95% intervals after averaging three seeds within each of 14 people. The direct-coordinate row is jointly trained; the three repair rows share a frozen encoder. C–D: paired comparator-minus-candidate waveform gains using the same person- $t$  procedure, positive favoring the first named arm. Delta dense versus low scalar is the declared primary; endpoint is secondary. Other displayed intervals are exploratory and unadjusted, and are conditional on the three fitted seeds. All waveform scores include 180° failures. Marginal intervals in A–B do not substitute for paired contrasts in C–D.

#### 4.5 RESPONSE RANKING DOES NOT ESTABLISH ANATOMICAL NAMING FIDELITY

Laterality motivated the signed response, but geometric naming was examined post hoc. The diagnostic admits reference-valid left/right hip, knee, and ankle pairs separated by at least four pixels. Let  $d_n, d_s$  be summed Euclidean errors under named and swapped correspondence. A prediction fails if swapped assignment is better by more than two pixels, the assignments are within two pixels (ambiguous), or predicted coordinates are nonfinite (missing). This denominator differs from angular support. The combined export does not permit a verified component breakdown, and 50% is not an established chance level.

Under global naming swaps, assignment failure reaches 81.12% for endpoint and 83.40% for delta, compared with 23.29% for direct coordinate fitting (Figure 5). The pooled delta rate is 50.46%, versus endpoint’s 49.69%, despite delta’s lower response-error point estimate. The exploratory excess is 0.77 percentage points [0.17, 1.53] (crossed 95% interval), unadjusted and not a replacement for the response primary. This empirical disagreement shows that the response ranking does not establish side recovery in these fitted outputs. Direct is not universally best: its left/right excursion errors are 17.94/17.19°, versus delta’s 15.70/15.29°, including 180° failure costs. Different outcomes answer different restoration questions.

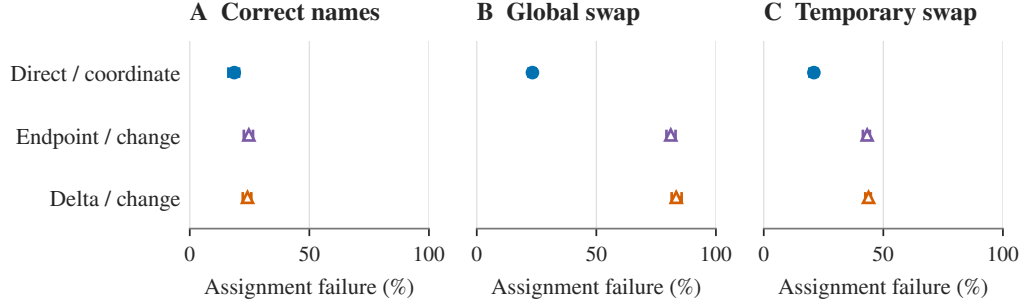


Figure 5: Post hoc geometric assignment diagnostic. Points show failure means for every naming condition; intervals are exploratory crossed-person/seed 95% intervals over 14 people and three seeds. Each estimates its method mean, not a pairwise effect. Failure combines wrong, ambiguous, and missing predictions on separately eligible bilateral hip/knee/ankle pairs. Endpoint strata retain the executed 4:1 nonheld/held weighting. All estimators, cameras, physical orientations, and observation conditions are pooled. Direct uses joint coordinate fitting; endpoint and delta use frozen encoders and the original change readout. No categorical lines or chance baseline are implied.

## 5 DISCUSSION AND LIMITATIONS

The comparison suggests a practical standard for evidence about predictive representations in restoration: a feature objective should be evaluated through its decoded trajectories, the movement measurements derived from them, and the reliability of the measurements. In this study, the zero-response benchmark changes the meaning of improvement over noisy poses. Failure accounting shows that rare invalid outputs materially affect the apparent feature advantage, while the naming diagnostic reveals a separate discrepancy in the same procedures. Together, these measurements distinguish a lower response score from more faithful restoration in the fitted systems.

The representation result remains open for specific reasons. Its primary interval spans both benefit and harm, the coordinate-readout ordering reverses, and equal auxiliary coefficients produce very unequal initial influence. Direct fitting is a useful practical comparator but has different encoder adaptation and coordinate-target exposure. The repair demonstrates a substantial change in waveform error after reducing scalar weight; it does not establish a privileged mechanism or a general case against change supervision. These distinctions identify what the completed evidence can adjudicate and what a better-controlled comparison would need to vary.

The main population limitation is the same 14 repeatedly inspected development people from two collections. Three fitted seeds provide coarse information about training variability, and new strata or bootstrap draws do not supply independent replications. There is no completed protected-person confirmation, GAVD evaluation, or natural-video study with independent anatomical references. Synthetic knee edits do not reproduce disease or treatment response; 2D angles are camera-dependent. Independent biomechanical validation, as exemplified by systems such as OpenCap (Uhlrich et al., 2023), addresses a different evidential requirement from exact agreement with a chosen synthetic projection.

Optimization and comparators also limit generalization. The study uses fixed budgets, a compact residual readout, and a narrow set of calibrated weights rather than a convergence or tuning sweep. Separate teachers and the absence of a new movement-pair randomization control prevent unique attribution to feature-difference coupling. External temporal refiners and masked-motion systems were not trained here. Consequently, these outcomes characterize the implemented training procedures rather than the best attainable performance of their method families.

The accompanying source package retains configurations, ledgers, person-level results, condition exports, analysis scripts, and vector figure sources. These support recalculating the reported summaries and intervals. The compact local packet lacks the raw rendered frames, per-example reconstructed coordinates, and full checkpoints needed for an end-to-end rerun or verified qualitative

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486 reconstruction panels. Tutorial fixtures are excluded from scientific results. The appendix provides  
487 a reconstruction contract and complete result inventories, while the claim-to-artifact audit separates  
488 reanalysis from new experiments.

489 A decisive follow-up would freeze the paired response and waveform comparisons before testing  
490 uninspected people, compare matched auxiliary influence across weights and budgets, and retain  
491 zero-response and jointly trained coordinate controls. Anatomical naming should be evaluated  
492 prospectively under natural observations with synchronized references. Until those tests are avail-  
493 able, the supported conclusion is that supervision and reliability materially shape apparent restora-  
494 tion gains, and the incremental benefit of the tested feature-prediction procedures remains unre-  
495 solved.

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## AI USE DISCLOSURE

AI assistants helped inspect source code, notebooks, and saved experiment exports; formulate and critique the scientific framing; verify primary literature and submission instructions; draft and revise the manuscript; create plotting, analysis, and document-build code; and inspect numerical summaries and rendered pages. Independent assistant reviews covered scientific contribution, methods, evidence, statistics, writing, and figures. They did not recruit participants, collect source data, run new source training experiments, or provide independent empirical validation. Human authors remain responsible for verifying the complete manuscript, references, disclosure, and submission metadata before submission; this document does not assert that such verification has occurred.

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## A IMPLEMENTATION AND RECONSTRUCTION CONTRACT

This appendix describes the completed procedure, not a new experimental run. The retained configuration and ledgers identify what was fitted. Current source defaults alone are not used to infer executed settings. All manuscript estimates come from the saved development exports; synthetic tutorial examples and small notebook fits are excluded.

### A.1 MOVEMENT GENERATION, OBSERVATIONS, AND NORMALIZATION

The source selects walking candidates from the eligible AMASS manifest, partitions nonoverlapping 128-frame windows, and screens projected geometry. Filename selection and geometric admission do not independently verify clinical status or natural gait for every interval. The local right-knee rotation is composed with recorded body-model motion, gated by an ankle-height proxy, and tapered to vanish at window boundaries. Edit levels are 0, 5, 10, and 15 degrees. Editing precedes reflection about a pelvis-defined plane and exchange of body-model side indices. Cameras are fitted once across the original, edited, and mirrored geometries of each source window, then retained. Oblique and side views use 45 and 90 degrees. The nominal rotation is not a claim about the resulting 2D angle change.

Rendered person boxes are supplied to the fixed estimators. The selected joints, in bilateral pairs, are shoulders, elbows, wrists, hips, knees, and ankles. Naming corruption permutes the coordinate, confidence, and availability channels together; timestamps and anatomically labeled references are unchanged. Global swaps affect the full sequence; temporary swaps cover its middle third. Image occlusion and artificial pretraining masks are different operations. Reference coordinates are projected body-model joints, including joints hidden in an image. They are exact for this chosen model and projection, not independently measured anatomy.

For endpoint  $i$ , let  $O_i(t, j)$  be observation availability and  $M_i(t, j)$  the artificial query mask repeated over four-frame patches. Context is  $C_i = O_i \wedge \neg M_i$ . Normalization uses only context coordinates:

$$o_i = \text{median}(X_{C_i}), \quad s_i = \|P_{95}(X_{C_i}) - P_5(X_{C_i})\|_2, \quad \tilde{X}_i = (X_i - o_i)/s_i. \quad (4)$$

Medians and percentiles operate separately on the two axes. With fewer than two context points, a nonfinite span, or span below  $10^{-6}$ , the recorded fallback is  $o_i = (0, 0)$ ,  $s_i = 1$ . References and hidden input coordinates cannot affect this transform. Teacher references use the same endpoint-specific observation transform; predictions return to pixels by  $\hat{Y}_i = s_i \hat{Y}_i^{\text{norm}} + o_i$ . Inference adds no artificial mask.

### A.2 TOKENS, MASKING, AND READOUT

Each four-frame joint patch contains five channels per frame: safe coordinates, confidence, availability, and seconds since the first timestamp. A linear  $20 \rightarrow 96$  map, learned joint embedding, and sinusoidal time position form 384 tokens. The encoder has four layers and four attention heads; the predictor has two layers. All slots remain in the bidirectional sequence. Masking zeros coordinate, confidence, and availability channels; time and positional identifiers remain. Tokens with no usable frame additionally receive a learned missing-query embedding.

Graph-time masking selects connected joint regions and cyclic time intervals. For  $n > 1$  available tokens, it fills the exact budget  $\min(\max(1, \text{round}(0.5n)), n - 1)$ ; otherwise no artificial mask is added. A token is available if any frame is observed. Region overlap and final-region truncation are retained in the mask audit. Alternative mask policies exist in code but are not completed experimental comparisons in this paper.

The residual readout is LayerNorm, linear (96, 96), GELU, and linear (96, 8), with the last layer initialized to zero. It adds residuals at observed positions and emits absolute normalized coordinates at missing positions. Frozen readouts use the retained online encoder, discarding teacher and predictor. Their fresh seed-dependent initialization is shared across comparison arms using seed +100003; optimizers then evolve independently. Coordinate pretraining’s head is replaced by this fresh readout. Direct fitting updates encoder and readout together.

### A.3 COORDINATE AND BASE FEATURE OBJECTIVES

Final coordinate fitting uses reference validity  $V_i$  and supported endpoints  $I$ :

$$L_x = \frac{1}{|I|} \sum_{i \in I} \frac{\sum_{t,j} V_i(t,j) \|\hat{Y}_{i,t,j}^{\text{norm}} - Y_{i,t,j}^{\text{norm}}\|_2^2}{2 \sum_{t,j} V_i(t,j)}. \quad (5)$$

Masked coordinate pretraining first averages valid frames within queried joint/patch tokens, then supported tokens within endpoints, then endpoints equally. This squared normalized loss differs from evaluation NLE, which uses unsquared Euclidean pixel distance divided by the reference rendering-box diagonal.

Write  $p_{ik}$  for student predictor logits and  $t_{ik}$  for detached teacher logits. To avoid conflating logits with probabilities, let  $Q_{ik} = \text{softmax}((t_{ik} - c)/0.06)$  and  $P_{ik} = \text{softmax}(p_{ik}/0.1)$ . The base feature loss is

$$L_{\text{base}} = \text{mean}_i \text{mean}_{k \in Q_i} [-Q_{ik}^\top \log P_{ik}] + 0.05 L_{\text{VICReg}}. \quad (6)$$

Queries satisfy artificial hiding or any missing input frame, with at least one valid reference frame in the token. Teacher input comprises projected reference coordinates, reference validity as confidence/availability, and physical timestamps. This centered cross-entropy mechanism follows the relevant S-JEPA/DINO precedent; it is not I-JEPA feature regression.

VICReg is applied to projector outputs from the mean student token features in two independently translated input views. Each translation coordinate is uniform in  $[-0.02, 0.02]$  normalized units. The loss is  $25L_{\text{inv}} + 25L_{\text{var}} + L_{\text{cov}}$ : mean squared view discrepancy; average  $\text{ReLU}(1 - \sqrt{\text{Var}} + 10^{-4})$  with population variance; and squared off-diagonal sample covariances, averaged over views and divided by feature dimension. Singleton supported batches cannot supply this regularizer.

For Equation 2, common auxiliary support requires queries in both states and all four reference frames valid in both patches. Supported tokens are averaged within each pair and pairs equally. An unsupported auxiliary pair does not erase supported base supervision. On identical tensors,  $L_{\Delta,k} - L_{E,k} = -e_a^\top e_b / D$ , but separately fitted teachers evolve differently. Independent endpoint masks also produce distinct observation normalizations; the feature difference is not a calibrated physical unit. The shuffled-reference control selects a different accepted window from the same person while retaining endpoint role and observation conditions. It does not randomize pairs inside the new feature-difference auxiliary.

The coordinate-delta control uses  $s_{ab} = (s_a + s_b)/2$  and  $r_i = (s_i/s_{ab})(\hat{Y}_i^{\text{norm}} - Y_i^{\text{norm}})$ . Its per-frame loss is  $\|r_b - r_a\|^2/4$ , then averaged over four-frame patches, common supported tokens, and pairs. Its separate coefficient is  $\lambda_C = 0.1 \sqrt{G_{\text{coordinate base}}/G_{\text{coordinate delta}}} = 3.79389230231159$ , with gradient energies defined as in Section A.5.

### A.4 ANGULAR LOSSES AND REPAIR

For knee-to-hip and knee-to-ankle vectors  $u, v$ , the projected angle is

$$\theta = \text{atan2}(|u_x v_y - u_y v_x|, u^\top v) 180/\pi. \quad (7)$$

Percentiles use linear interpolation. The implementation guards  $\text{atan2}(0, 0)$ ; an exactly straight knee has a zero subgradient through the absolute cross product. Training support  $S_{ab}$  is determined from common reference validity for both legs and states, reference segments at least two pixels, and at least 16 frames and 80% of the window. Predictions cannot remove that support. With  $\Delta$  denoting state  $b - a$ ,

$$L_s = \mathbb{E}_{(a,b)} \left[ \left( \frac{\Delta \hat{A} - \Delta A}{180} \right)^2 \right], \quad L_d = \mathbb{E}_{(a,b)} \left[ \frac{1}{|S_{ab}|} \sum_{(t,\ell) \in S_{ab}} \left( \frac{\Delta \hat{\theta}_{t,\ell} - \Delta \theta_{t,\ell}}{180} \right)^2 \right]. \quad (8)$$

The geometry loss averages  $\text{ReLU}((2 - \text{length})/2)^2$  over predicted segments, legs, endpoints, and common frames, then pairs equally. The original readout is  $L_x + L_s + L_g$ ; low scalar is  $L_x + 0.1L_s + L_g$ ; dense is  $L_x + \lambda_d L_d + L_g$ .

For each retained encoder and seed, 32 training-only batches at an identical fresh readout determine

$$\lambda_d = 0.1 \sqrt{\frac{\sum_b \|\nabla_\phi L_s^{(b)}\|_2^2}{\sum_b \|\nabla_\phi L_d^{(b)}\|_2^2}}, \quad (9)$$

where  $\phi$  includes every readout parameter and unused gradients count as zero. Thus dense is initially matched to low scalar, not to coordinate loss. Encoder and teacher remain frozen; there is no readout-stage teacher update. Equal initial magnitude does not match the later optimization path or gradient directions.

#### A.5 OPTIMIZATION AND CALIBRATION AUDIT

Pretraining and frozen readout fitting each use 2,000 updates; direct fitting uses 4,000. Batch size is 16. Training samples person, source motion, window, and accepted pair uniformly in succession. AdamW uses peak learning rate  $3 \times 10^{-4}$ , betas (0.9, 0.999), epsilon  $10^{-8}$ , weight decay 0.01, and global gradient-norm clipping at one. For  $U$  updates,  $k = 0, \dots, U - 1$ , and  $W = \max(1, \text{round}(0.05U))$ , the rate is

$$\eta_k = \begin{cases} \eta_0(k+1)/W, & k < W, \\ \frac{\eta_0}{2} [1 + \cos(\pi(k-W)/(U-W))], & k \geq W. \end{cases} \quad (10)$$

After each feature-pretraining student update, teacher parameters use EMA momentum  $m_k = 0.999 - 0.009[1 + \cos(\pi k/(U-1))]/2$ , increasing from 0.99 to 0.999. The center then updates as  $c \leftarrow 0.9c + 0.1 \text{mean}_i \text{mean}_{k \in \mathcal{Q}_i} t_{ik}$ , using pre-update teacher tokens with equal supported-endpoint weighting.

For feature calibration, let  $G_j = \sum_{b=1}^{32} \|\nabla_\psi L_j^{(b)}\|_2^2$  over all  $P = 702,624$  trainable coordinates, and  $R_j = \sqrt{G_j/(32P)}$ . At seed-17 initialization, with no parameter updates,

$$\lambda_J = 0.1 \sqrt{G_{\text{base}} / \max(G_\Delta, G_E)} = 0.012646811176583523. \quad (11)$$

The same coefficient is reused across all fitted seeds. Table 3 reports its initial influence, measured before clipping. It is an RMS gradient-magnitude ratio, not a squared-energy ratio or a component-specific AdamW update.

Table 3: Verified feature calibration at initialization. The percentage is  $100\lambda_J R_j / R_{\text{base}}$  for each auxiliary. Source: the retained loss-calibration receipt; all eight bound source-code hashes match the audited [checkout](#).

| Component          | $G_j$            | $R_j$         | Weighted % of base |
|--------------------|------------------|---------------|--------------------|
| Base feature loss  | 90,192.748779    | 0.06333581046 | 100 (reference)    |
| Endpoint auxiliary | 5,639,096.859375 | 0.50080458681 | 10.000000          |
| Delta auxiliary    | 0.095846         | 0.00006529059 | 0.001303714        |

The six new JEPA encoders clip 11,999 of 12,000 pretraining updates: delta seeds 17/29/43 clip 2,000/2,000/1,999; endpoint seeds clip 2,000 each. Coordinate-delta encoders clip zero of their 6,000 updates. These are ledger counts of combined-gradient rescaling, not diagnoses of convergence or persistent component influence. The transferred packet does not include a complete gradient time series, matched-influence retraining, or a convergence sweep.

## B COMPLETED DESIGN, MEASUREMENT, AND INFERENCE

The training population contains 112 people and 1,645 windows from 692 raw motions. The development population contains 14 people and 155 windows; 12 people are from BioMotionLab\_NTroje and two from KIT. Repeated conditions give 55,800 endpoint records, including a duplicate zero-level endpoint for the no-change diagnostic. Nonzero paired-response scoring excludes the zero intervention. The 128 samples at 25 Hz span 5.08 seconds from first to last sample.

Table 4: Completed fits across the three sequential stages. Counts include all three seeds, not independent populations. Repairs reuse the six retained feature encoders. The larger 102-fit tutorial recipe is not executed evidence.

| Stage            | Pretrains | Final fits | Scope                                                        |
|------------------|-----------|------------|--------------------------------------------------------------|
| Core             | 9         | 30         | Five families, coordinate/change; six deterministic controls |
| Feature response | 9         | 18         | Delta/endpoint features and coordinate delta; two readouts   |
| Readout repair   | 0         | 12         | Two retained encoders; low scalar/dense; six calibrations    |

The common angular reference mask intersects support across every variant in a source family. Eligible measurements require finite reference geometry, reference segment lengths at least two pixels, at least 16 supported frames, and 80% coverage. Failed predictions remain in eligible denominators. Costs are  $180^\circ$  for waveform and per-leg excursion,  $360^\circ$  for asymmetry  $A$ ,  $720^\circ$  for movement/nuisance response, and  $1,440^\circ$  for the higher-order interaction. These are outcome-specific costs, not a universal angular threshold. Direction accuracy requires  $|\Delta A| > 1^\circ$  and counts failures as wrong; a 50% score has no calibrated chance interpretation without its reference-sign distribution.

Geometric assignment is separately evaluated on reference-valid bilateral hip, knee, and ankle pairs with reference separation at least four pixels. Summed pointwise Euclidean error under the named versus swapped correspondence determines wrong or ambiguous assignment at a two-pixel margin; nonfinite prediction is missing. All three count as failure. The compact packet stores their combined rate, so the manuscript does not invent a component decomposition.

Aggregation first averages conditions within each window, then windows within raw motions, motions within people, and fitted seeds. In condition reaggregation, endpoint metrics weight non-held/held strata 4:1 because of baseline endpoint reuse and the duplicated zero endpoint; nonzero response metrics use 2:1. An unweighted average of exported strata would change the original estimand. All newly displayed intervals retain 14 people and the three seeds, rather than counting windows, renderings, or 42 person-seed cells as independent participants.

Crossed bootstraps use 2,000 draws, RNG seed 731, independently resampling the 14 people and three seeds while keeping methods paired within sampled cells. The repair primary instead averages seeds within each person and uses  $\bar{d} \pm t_{.975,13} s_d / \sqrt{14}$ . This is conditional on the fitted seeds. Repair’s crossed interval is a sensitivity analysis. New exploratory mean intervals follow the procedure stated in their captions; their overlap or separation is not a replacement for a paired-effect interval.

Local configurations predate their corresponding ledgers, but creation timestamps and hashes are not independent preregistration records. The core, response, and repair stages reuse the development population and follow inspection of earlier results. The response primary remains delta versus endpoint with the original change readout; coordinate readout is secondary. The repair primary remains delta dense versus low scalar on ViTPose waveform error. Geometric assignment, the new strata, penalty sensitivities, and expanded figure intervals are post hoc, unadjusted development analyses. A protected-person confirmation has no completed result.

## C COMPLETE POOLED RESPONSE-STUDY INVENTORY

Table 5 includes every learned variant in the pooled response export. It is a complete inventory, not a significance ranking. Direct methods jointly fit encoder and readout; all remaining families freeze the encoder before readout training. “Coordinate” names the readout objective; “change” names  $L_x + L_s + L_g$ . The coordinate-pretrained family is distinct from the coordinate readout objective.

Table 5: All 16 neural variants. Response and waveform are in degrees, NLE is dimensionless, Fail is the weighted response failure percentage, and Assign is geometric assignment failure percentage. Means use 14 people and three seeds. All are lower-is-better outcomes, but they have different support and cannot be combined into an unspecified overall ranking.

| Method / objective            | Response | Waveform | NLE    | Fail (%) | Assign (%) |
|-------------------------------|----------|----------|--------|----------|------------|
| Direct / coordinate           | 7.54     | 12.07    | 0.0298 | 0.344    | 20.95      |
| Direct / change               | 10.75    | 19.28    | 0.0518 | 0.711    | 39.39      |
| Initialized / coordinate      | 9.43     | 16.97    | 0.0687 | 0.388    | 50.83      |
| Initialized / change          | 10.16    | 21.96    | 0.0718 | 0.525    | 49.70      |
| Shuffled JEPA / coordinate    | 9.35     | 16.74    | 0.0676 | 0.390    | 50.29      |
| Shuffled JEPA / change        | 11.75    | 21.37    | 0.0718 | 0.754    | 49.49      |
| Coordinate / coordinate       | 10.25    | 15.88    | 0.0487 | 0.571    | 39.08      |
| Coordinate / change           | 11.17    | 21.40    | 0.0561 | 0.711    | 44.48      |
| Core JEPA / coordinate        | 10.69    | 17.45    | 0.0626 | 0.613    | 48.64      |
| Core JEPA / change            | 10.07    | 22.55    | 0.0700 | 0.539    | 49.70      |
| Coordinate delta / coordinate | 10.18    | 16.22    | 0.0550 | 0.560    | 42.92      |
| Coordinate delta / change     | 14.52    | 21.08    | 0.0617 | 1.170    | 47.75      |
| Endpoint JEPA / coordinate    | 10.23    | 16.90    | 0.0632 | 0.526    | 49.97      |
| Endpoint JEPA / change        | 10.36    | 22.03    | 0.0704 | 0.563    | 49.69      |
| Delta JEPA / coordinate       | 10.94    | 17.38    | 0.0628 | 0.647    | 48.98      |
| Delta JEPA / change           | 9.99     | 21.76    | 0.0699 | 0.525    | 50.46      |

Unchanged observations have response error  $12.688^\circ$ , waveform error  $18.571^\circ$ , and all-joint NLE 0.071772. Direct coordinate fitting gives  $7.544^\circ$ ,  $12.073^\circ$ , and 0.029791. Zero response has mean error  $5.810830^\circ$  and no waveform, coordinates, or anatomical naming prediction. Assigning it zero waveform or coordinate error would invent an outcome it does not produce.

The original package increases waveform means across all eight families. The five core families show that direction for all 14 person means and all three seed means; this is descriptive consistency within one reused cohort. The minimum family mean increase across all eight is approximately  $4.39^\circ$ . Core JEPA’s response improves while its waveform worsens, illustrating the empirical trade-off without claiming the scalar term alone caused it.

The three declared primary effects remain unresolved: core JEPA versus direct under change,  $0.688^\circ$   $[-0.640, 1.977]^\circ$  response improvement; delta versus endpoint under change,  $0.373^\circ$   $[-1.110, 1.760]^\circ$  response improvement; and delta dense versus low scalar,  $0.279^\circ$   $[-0.195, 0.753]^\circ$  ViTPose waveform improvement. The first two use crossed intervals; the third uses the person- $t$  procedure. No pooled meta-effect is implied.

## D ALL SELECTED OBSERVATION STRATA

These analyses are exploratory. Tables 6 and 7 retain all selected groups and all 16 neural variants. Every group contains the same 14 people and three fitted seeds. Nominal edit groups use seen 5/10° levels versus held 15°; these are not bins of actual projected response magnitude. The actual zero-response mean for each group supplies its reference-response scale. The files retain method means and paired effects with crossed intervals; the compact tables show means to keep the full inventory legible.

Table 6: Response error (°) in all observation-by-nominal-edit groups, retaining the original 720° failure cost. Clear/occluded describe image observations. The 15° edit was excluded from fitting.

| Method / objective            | Clear 5/10 | Clear 15 | Occl. 5/10 | Occl. 15 |
|-------------------------------|------------|----------|------------|----------|
| Zero response                 | 4.10       | 9.23     | 4.10       | 9.23     |
| Direct / coordinate           | 2.79       | 6.12     | 9.54       | 14.49    |
| Direct / change               | 3.67       | 7.87     | 14.89      | 19.53    |
| Initialized / coordinate      | 4.76       | 10.40    | 10.62      | 15.44    |
| Initialized / change          | 4.67       | 10.10    | 12.18      | 17.14    |
| Shuffled JEPA / coordinate    | 4.66       | 10.24    | 10.63      | 15.25    |
| Shuffled JEPA / change        | 5.35       | 10.87    | 14.69      | 19.54    |
| Coordinate / coordinate       | 4.15       | 9.24     | 13.08      | 17.78    |
| Coordinate / change           | 4.19       | 9.38     | 14.87      | 19.56    |
| Core JEPA / coordinate        | 4.37       | 9.73     | 13.54      | 18.61    |
| Core JEPA / change            | 4.32       | 9.64     | 12.63      | 16.86    |
| Coordinate delta / coordinate | 4.18       | 9.32     | 12.60      | 18.19    |
| Coordinate delta / change     | 4.27       | 9.48     | 21.33      | 26.44    |
| Endpoint JEPA / coordinate    | 4.53       | 9.98     | 12.39      | 17.57    |
| Endpoint JEPA / change        | 4.54       | 9.95     | 12.65      | 17.82    |
| Delta JEPA / coordinate       | 4.37       | 9.73     | 14.05      | 19.09    |
| Delta JEPA / change           | 4.33       | 9.64     | 12.08      | 17.47    |

Table 7: Response error (°) in all observation-by-estimator groups, with 2:1 seen/hold nonzero-response weights. R: RTMPose-M; H: HRNet-W32; V: ViTPose-base. ViTPose was excluded from fitting. Original failure costs remain.

| Method / objective            | Clear R | Clear H | Clear V | Occl. R | Occl. H | Occl. V |
|-------------------------------|---------|---------|---------|---------|---------|---------|
| Zero response                 | 5.81    | 5.81    | 5.81    | 5.81    | 5.81    | 5.81    |
| Direct / coordinate           | 4.07    | 3.85    | 3.77    | 6.37    | 10.18   | 17.02   |
| Direct / change               | 4.94    | 5.33    | 4.95    | 9.06    | 16.49   | 23.75   |
| Initialized / coordinate      | 6.62    | 6.74    | 6.57    | 7.90    | 11.06   | 17.71   |
| Initialized / change          | 6.34    | 6.73    | 6.38    | 9.53    | 13.09   | 18.89   |
| Shuffled JEPA / coordinate    | 6.50    | 6.62    | 6.44    | 7.38    | 12.75   | 16.39   |
| Shuffled JEPA / change        | 7.36    | 7.11    | 7.09    | 10.80   | 18.54   | 19.58   |
| Coordinate / coordinate       | 5.84    | 5.87    | 5.84    | 8.04    | 13.68   | 22.22   |
| Coordinate / change           | 5.89    | 5.95    | 5.91    | 8.17    | 19.30   | 21.82   |
| Core JEPA / coordinate        | 6.06    | 6.28    | 6.11    | 7.09    | 17.07   | 21.52   |
| Core JEPA / change            | 6.06    | 6.14    | 6.06    | 9.66    | 14.02   | 18.44   |
| Coordinate delta / coordinate | 5.89    | 5.90    | 5.89    | 8.35    | 14.52   | 20.52   |
| Coordinate delta / change     | 5.96    | 6.11    | 5.95    | 11.72   | 25.19   | 32.19   |
| Endpoint JEPA / coordinate    | 6.28    | 6.46    | 6.29    | 7.81    | 16.32   | 18.22   |
| Endpoint JEPA / change        | 6.23    | 6.54    | 6.26    | 10.53   | 14.79   | 17.81   |
| Delta JEPA / coordinate       | 6.08    | 6.29    | 6.10    | 7.62    | 16.57   | 22.99   |
| Delta JEPA / change           | 6.03    | 6.17    | 6.10    | 8.72    | 15.16   | 17.75   |

The compact `response-curves-person.csv` contains signed reference responses already averaged over source windows, motions, and nuisance factors. In general,  $|\mathbb{E}[\Delta A]| \neq \mathbb{E}[|\Delta A|]$ . It therefore cannot reconstruct per-pair absolute-magnitude bins or their counts. We do not use absolute averaged responses as a substitute, and do not infer the individual response distribution from its pooled zero-response mean.

## E FAILURE ACCOUNTING AND ALL FIXED-COST SENSITIVITIES

Equation 3 uses the same population weights for successful-error and failure-cost contributions. Table 8 keeps both distinct from a conditional error. If  $U$  is the unconditional successful contribution and  $f = \Pr_w(F)$ , the reported conditional column is  $U/(1 - f)$ . This conditions the original weighted population on success and can change effective person weights. It is not the separately exported average of successful source-family means, which applies conditioning before the motion/person hierarchy. Both are retained in the audit CSV; neither is directly additive with failure cost.

Table 8: Response reliability for all variants. Fail is a person-balanced percentage. Success is  $U$ , Cost is  $720f$ , Total is their sum, and Conditional is  $U/(1 - f)$ , all in degrees. Raw failed-pair counts across seeds are retained in the audit and are not substituted for weighted rates.

| Method / objective            | Fail (%) | Success | Cost  | Total  | Conditional |
|-------------------------------|----------|---------|-------|--------|-------------|
| Direct / coordinate           | 0.344    | 5.069   | 2.475 | 7.544  | 5.087       |
| Direct / change               | 0.711    | 5.634   | 5.120 | 10.754 | 5.674       |
| Initialized / coordinate      | 0.388    | 6.642   | 2.791 | 9.433  | 6.668       |
| Initialized / change          | 0.525    | 6.380   | 3.779 | 10.159 | 6.414       |
| Shuffled JEPA / coordinate    | 0.390    | 6.539   | 2.806 | 9.345  | 6.565       |
| Shuffled JEPA / change        | 0.754    | 6.319   | 5.427 | 11.746 | 6.367       |
| Coordinate / coordinate       | 0.571    | 6.137   | 4.110 | 10.247 | 6.173       |
| Coordinate / change           | 0.711    | 6.059   | 5.116 | 11.174 | 6.102       |
| Core JEPA / coordinate        | 0.613    | 6.280   | 4.411 | 10.691 | 6.319       |
| Core JEPA / change            | 0.539    | 6.186   | 3.881 | 10.066 | 6.219       |
| Coordinate delta / coordinate | 0.560    | 6.145   | 4.032 | 10.177 | 6.180       |
| Coordinate delta / change     | 1.170    | 6.094   | 8.426 | 14.520 | 6.166       |
| Endpoint JEPA / coordinate    | 0.526    | 6.443   | 3.790 | 10.233 | 6.477       |
| Endpoint JEPA / change        | 0.563    | 6.306   | 4.055 | 10.361 | 6.342       |
| Delta JEPA / coordinate       | 0.647    | 6.287   | 4.658 | 10.945 | 6.328       |
| Delta JEPA / change           | 0.525    | 6.210   | 3.778 | 9.988  | 6.242       |

For the primary feature comparison, the failure-cost difference is  $4.055115 - 3.778439 = 0.276675^\circ$ , versus a total difference of  $0.373054^\circ$ . Its 74.17% share is arithmetic score accounting. The weighted endpoint/delta failure rates are 0.563210%/0.524783%; these must not be confused with the share of a mean contrast. Direct coordinate fitting has weighted failure rate 0.343762%, unconditional successful contribution  $5.069067^\circ$ , and failure contribution  $2.475088^\circ$ .

At costs 0/180/360/720°, paired endpoint-minus-delta effects are respectively 0.0964 [−0.0226, 0.2393], 0.1655 [−0.1767, 0.5269], 0.2347 [−0.4729, 0.9413], and 0.3731 [−1.1103, 1.7603] degrees, with crossed 95% intervals. The zero-cost value is not a success-conditional error; it assigns zero loss to failed cases. Table 9 reports the complete fixed-cost sweep rather than selecting the cost that produces a preferred ranking.

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Table 9: All rescored response means ( $^{\circ}$ ) with unchanged predictions and failure indicators. Costs 180 and 360 are quarter/half-cost stress tests; zero is an accounting lower bound; 720 is the original maximum valid-response-error bound. All new sensitivities are exploratory.

| Method / objective            | Cost 0 | Cost 180 | Cost 360 | Cost 720 |
|-------------------------------|--------|----------|----------|----------|
| Zero response                 | 5.811  | 5.811    | 5.811    | 5.811    |
| Direct / coordinate           | 5.069  | 5.688    | 6.307    | 7.544    |
| Direct / change               | 5.634  | 6.914    | 8.194    | 10.754   |
| Initialized / coordinate      | 6.642  | 7.340    | 8.038    | 9.433    |
| Initialized / change          | 6.380  | 7.325    | 8.270    | 10.159   |
| Shuffled JEPa / coordinate    | 6.539  | 7.241    | 7.942    | 9.345    |
| Shuffled JEPa / change        | 6.319  | 7.676    | 9.032    | 11.746   |
| Coordinate / coordinate       | 6.137  | 7.165    | 8.192    | 10.247   |
| Coordinate / change           | 6.059  | 7.338    | 8.616    | 11.174   |
| Core JEPa / coordinate        | 6.280  | 7.383    | 8.486    | 10.691   |
| Core JEPa / change            | 6.186  | 7.156    | 8.126    | 10.066   |
| Coordinate delta / coordinate | 6.145  | 7.153    | 8.161    | 10.177   |
| Coordinate delta / change     | 6.094  | 8.201    | 10.307   | 14.520   |
| Endpoint JEPa / coordinate    | 6.443  | 7.390    | 8.338    | 10.233   |
| Endpoint JEPa / change        | 6.306  | 7.320    | 8.334    | 10.361   |
| Delta JEPa / coordinate       | 6.287  | 7.451    | 8.616    | 10.945   |
| Delta JEPa / change           | 6.210  | 7.154    | 8.099    | 9.988    |

## F REPAIR INVENTORY AND INTERPRETATION

The repair follows inspection of earlier development results. It retains the original feature encoders, trains new readouts, and compares original change, low scalar, and dense supervision without changing the coordinate and geometry terms. Coordinate-readout and direct references remain in the full export. The declared primary is the delta encoder’s dense-versus-low-scalar ViTPose waveform contrast; a favorable endpoint secondary result cannot replace it.

Table 10: Every retained ViTPose method in the repair export. Waveform and response are degrees; NLE is dimensionless; response failures are weighted percentages. Direct is jointly fitted; all other methods use frozen encoders. This table preserves the coordinate-only references in addition to the three plotted repair arms.

| Method / supervision        | Waveform | Response | NLE    | Response fail (%) |
|-----------------------------|----------|----------|--------|-------------------|
| Delta / coordinate          | 19.13    | 14.55    | 0.0652 | 1.123             |
| Delta / original change     | 23.17    | 11.93    | 0.0718 | 0.772             |
| Endpoint / coordinate       | 18.47    | 12.26    | 0.0657 | 0.786             |
| Endpoint / original change  | 23.33    | 12.04    | 0.0724 | 0.770             |
| Direct / coordinate (joint) | 13.02    | 10.40    | 0.0315 | 0.731             |
| Delta / dense               | 19.19    | 12.98    | 0.0669 | 0.917             |
| Delta / low scalar          | 19.47    | 13.11    | 0.0670 | 0.934             |
| Endpoint / dense            | 18.92    | 12.42    | 0.0671 | 0.830             |
| Endpoint / low scalar       | 19.64    | 12.32    | 0.0674 | 0.823             |

Delta’s original-to-low-scalar waveform gain is  $3.697397^\circ$  and its original-to-dense gain is  $3.976253^\circ$ . Their ratio is 92.98697%. It summarizes means and has no mechanistic or causal interpretation. The paired dense-versus-low difference is  $0.278856^\circ$ ; its person- $t$  interval is  $[-0.194914, 0.752625]^\circ$  and crossed sensitivity interval  $[-0.436648, 0.857563]^\circ$ . Ten people favor dense and four favor low scalar; seed-mean gains are  $-0.208729$ ,  $0.577246$ , and  $0.468051^\circ$ . Three seed means give coarse training-variation evidence.

Endpoint’s secondary dense-versus-low waveform gain is  $0.718163^\circ$   $[0.387453, 1.048872]^\circ$  under the person- $t$  procedure. Relative to the same encoder’s coordinate-only readout, dense changes delta waveform error by  $+0.068963^\circ$  and endpoint by  $+0.446150^\circ$  (higher is worse); the corresponding coordinate-minus-dense intervals span zero. Relative to direct coordinate fitting, dense waveform error is higher by  $6.172813^\circ$  for delta and  $5.895024^\circ$  for endpoint. These are differently adapted systems, not a matched representation ranking.

Dense-versus-low response gain for delta is  $0.129110^\circ$   $[-2.686467, 2.944686]^\circ$  by person- $t$ , with crossed sensitivity  $[-4.223610, 3.584143]^\circ$ . No noninferiority margin was specified or established. The waveform result therefore cannot be described as improving trajectories while demonstrably preserving response accuracy.

## G ARTIFACT MAP AND REPRODUCTION BOUNDARY

The source package includes the manuscript, bibliography, editable Matplotlib/SVG figures, shared style, generated tables, and executable audit. The source exports are outputs/iclr/walking-core, jepa-response, and readout-repair. Their person/condition CSVs, comparison JSONs, ledgers, and calibration receipts bind the estimates. Version-local files distinguish new exploratory analyses from saved primary comparisons. The audit checks complete person–seed panels, exact condition reaggregation, additive cost identities, and 16 retained comparison entries. Figures and tables read these verified exports directly and record provenance.

These commands regenerate summaries and documents, not the training study. An end-to-end re-run additionally requires the motion/body-model assets, renders, pose inputs, per-example predictions, and complete checkpoints absent from the compact packet. The schematic depicts mathematical relationships; no restored-pose examples are fabricated. Tutorial fits, the full planned matrix, protected-person plans, and GAVD inventories are excluded from scientific evidence.